Part C

Course title: SC7 Bayes Methods
Lecturer: Geoff Nicholls
Checked by: David Steinsaltz & External Examiners

Date of this version: 09-04-25

Do not turn this page until you are told that you may do so

1. Let $\pi(\theta),\ \theta \in \mathbb{R}^p$ be a prior for a parameter θ and let $p(y|\theta),\ y \in \mathbb{R}^n$ be an observation model. Let $S : \mathbb{R}^n \to \mathbb{R}^p$ be a p-dimensional summary statistic for data y. Denote by $y_{obs} \in \mathbb{R}^n$ the observed data and let $s_{obs} = S(y_{obs})$. For $s \in \mathbb{R}^p$ let $d(s, s_{obs})$ be a distance measure over $\mathbb{R}^p$ and for $\delta > 0$ let $\Delta_\delta(y_{obs}) = \{y \in \mathbb{R}^n : d(S(y), s_{obs}) < \delta\}$. Let $p(\Delta_\delta(y_{obs})|\theta) = \int_{\Delta_\delta(y_{obs})} p(y|\theta)dy$ and let $p(\Delta_\delta(y_{obs})) = \int_{\mathbb{R}^p} p(\Delta_\delta(y_{obs})|\theta)\pi(\theta)d\theta$.

   (a) [6 marks] Suppose $\theta \sim \pi(\cdot)$ and $Y \sim p(\cdot|\theta)$.

   (i) Write down the ABC posterior $\pi_{ABC}(\theta|y_{obs}) = \pi(\theta|Y \in \Delta_\delta(y_{obs}))$ in terms of the quantities defined above.

   (ii) Show that

$$\pi_{ABC}(\theta|y_{obs}) = \int_{\Delta_\delta(y_{obs})} \pi(\theta|y)p(y|Y \in \Delta_\delta(y_{obs}))dy\,,$$

   where $p(y|Y \in \Delta_\delta(y_{obs})) = p(y)\mathbb{I}_{y\in\Delta_\delta(y_{obs})}/p(\Delta_\delta(y_{obs}))$ and $p(y) = \int_{\mathbb{R}^p} p(y|\theta)\pi(\theta)d\theta$.

   (b) [10 marks] Consider a simple nonlinear climate model for convection rate (x_t) and horizontal (y_t) and vertical (z_t) temperature variation at time $t = 1, \ldots, n$ running over n days. For parameters $(\tau, \rho, \beta, \sigma) \in [0, \infty)^p$ with $p = 4$, the evolution equations are

$$\begin{aligned} x_{t+1} &= x_t + c\tau(y_t - x_t) + \epsilon_{1,t}\,, \\ y_{t+1} &= y_t + cx_t(\rho - z_t) - cy_t + \epsilon_{2,t}\,, \\ z_{t+1} &= z_t + cx_ty_t - c\beta z_t + \epsilon_{3,t}\,, \end{aligned}$$

   for $t = 1, \ldots, n-1$, with c a small fixed positive constant which is known and $\epsilon_{i,t} \sim N(0, \sigma^2)$ independent and identically distributed for $i = 1, 2, 3$ and $t = 1, \ldots, n-1$. Data $D_{obs} = (x_{obs}, y_{obs}, z_{obs})$ are gathered for a realisation of the process with $x_{obs} = (x_{obs,1}, \ldots, x_{obs,n})$ and similarly for y_{obs} and z_{obs}. These data are a direct realisation of the evolution equations above. It is of interest to estimate the parameters τ, ρ, β and σ from the data D_{obs}.

   (i) The data space is $\mathcal{D} = \mathbb{R}^{3n}$. Give a summary statistic $S : \mathcal{D} \to \mathbb{R}^p$ which might be used in an ABC algorithm aiming to approximate the posterior $\pi(\tau, \rho, \beta, \sigma|D_{obs})$.

   (ii) Let $D = (x, y, z)$ be a realisation of the process using some fixed parameters τ, ρ, β and σ and starting from $x_1 = x_{obs,1}, y_1 = y_{obs,1}$ and $z_1 = z_{obs,1}$. Explain how you would choose a suitable distance measure $d(S(D), S(D_{obs}))$ for use in ABC.

   (iii) Suppose a parameter prior $\pi(\tau, \rho, \beta, \sigma)$ on $[0, \infty)^p$ is given. You may assume this is easily simulated. Write down a rejection-ABC algorithm to sample $\pi_{ABC}(\tau, \rho, \beta, \sigma|D_{obs})$.

   (c) [9 marks] We now return to the generic setting set out at the start of this question. Let $\mu(y) = E_{\theta\sim\pi(\cdot|y)}(\theta)$ and let $|s|$ denote Euclidean norm in $\mathbb{R}^p$.

   (i) Suppose we are doing ABC using $S(y) = \mu(y)$ and $d(s, s_{obs}) = |s - s_{obs}|$. Show that

$$\left|E_{\theta\sim\pi_{ABC}(\cdot|y_{obs})}(\theta) - \mu(y_{obs})\right| < \delta.$$

   (ii) Let $f(y; w)$ be a function with fixed parameters w which takes input y and returns as output an estimate of $\mu(y)$. Let $\widehat{w}$ be an estimate for w chosen so that $f(y; \widehat{w}) \simeq \mu(y)$. Suppose that for $t = 1, \ldots, T$ we have samples $\theta_t \sim \pi(\cdot)$ and $y_t \sim p(\cdot|\theta_t)$ which are independent across pairs. Explain, and briefly justify, how you would use these samples to estimate w.

   (iii) Assemble these two results to give a generic rejection-ABC algorithm which learns its own summary statistics and uses them to do ABC.

2. Let $[L, U]$ be a finite interval and $\lambda : [L, U] \to \mathbb{R}^+$. In an inhomogeneous Poisson point process the probability for an event in a small interval δt is $\lambda(t)\delta t + o(\delta t)$ independently over intervals. If $Y = (Y_1, \ldots, Y_N)$ are the event times in $[L, U]$ then the sample space for Y is $\Omega = \cup_{n=0}^{\infty} \Omega_n$ where $\Omega_0 = \{\emptyset\}$ and

$$\Omega_n = \{y \in [L, U]^n : L < y_1 < y_2 < \cdots < y_n < U\}.$$

(a) [10 marks] The probability density to realise $Y = y,\ y = (y_1, \ldots, y_n)$ is

$$p(y|\lambda) = \exp(-\Lambda) \prod_{i=1}^{n} \lambda(y_i)\,,$$

where $\Lambda = \int_L^U \lambda(t)dt$.

(i) Let $Y \sim p(\cdot|\lambda)$ with $Y = (Y_1, \ldots, Y_N)$. Show that $N \sim \text{Poisson}(\Lambda)$.

(ii) Show that $p(y|\lambda)$ is normalised over Ω, that is $\Pr(Y \in \Omega|\lambda) = 1$.

(iii) Suppose $N = n$ is fixed. Consider an MCMC algorithm targeting $p(y|\lambda, n)$ at fixed $n > 1$. Let $y_0 = L$ and $y_{n+1} = U$. The proposal selects a component $i \sim U\{1, \ldots, n\}$ uniformly, samples $y_i' \sim U(y_{i-1}, y_{i+1})$ and proposes $y' = (y_1, \ldots, y_{i-1}, y_i', y_{i+1}, \ldots, y_n)$. Calculate the acceptance probability.

(b) [5 marks] For $j = 1, \ldots, J$ let $L < b_j < U$ be the birth year of individual j and let $\lambda(t)$ be the rate at which individuals catch a certain disease at time $L < t < U$. In a small interval δt an individual catches the disease with probability $\lambda(t)\delta t + o(\delta t)$ independently over time and people.

(i) Calculate the probability $p_j(\lambda)$ that individual j catches the disease at least once over the interval $[b_j, U]$.

(ii) A survey at time U tests J individuals alive at time U to see if they had caught the disease by that time. The survey records $z_j = 0$ if they never had the disease and $z_j = 1$ if they had it at least once. Let $z = (z_1, \ldots, z_J)$. Show that

$$p(z|\lambda) = \prod_{j=1}^{J} p_j(\lambda)^{z_j} (1 - p_j(\lambda))^{1-z_j}\,.$$

(c) [10 marks] Continuing the problem stated in Question 2(b), suppose for $j = 1, \ldots, J$ we have data $z_j \in \{0, 1\}$ and birth date b_j. We wish to infer $\lambda(t)$ for $L < t < U$. Let $\phi(t; a, b^2)$ be a normal density with mean a and variance b^2. Consider a normal mixture

$$\lambda(t) = \sum_{k=1}^{K} w_k \phi(t; \mu_k, \sigma_k^2),$$

with K components with means $L < \mu_k < U$, standard deviations $\sigma_k > 0$ and component weights $w_k > 0,\ k = 1, \ldots, K$. Let $\mu = (\mu_1, \ldots, \mu_K)$, $\sigma = (\sigma_1, \ldots, \sigma_K)$ and $w = (w_1, \ldots, w_K)$. Assume independent priors $\pi_\mu(\mu)$, $\pi_\sigma(\sigma)$ and $\pi_w(w)$ are given with $\pi_\theta(\theta) = \prod_k \pi_\theta(\theta_k)$ when θ is μ, σ or w. The prior for the number of components is $\pi_K(k) = \text{Poisson}(k; m)$ with $m > 0$ is a fixed prior hyperparameter.

(i) Write down the posterior $\pi(\mu, \sigma, w|z)$ (with $(b_1, \ldots, b_J)$ known and fixed) and give the space on which it is defined.

(ii) Give proposals and acceptance probabilities for a reversible jump MCMC algorithm targeting $\pi(\mu, \sigma, w|z)$ (just give updates adding and deleting single components).

3. (a) [12 marks] Let $\alpha > 0$ be given and let $H(dx) = h(x)dx$ be a continuous distribution with density $h(x)$ for $x \in \mathbb{R}^p$. Let $\theta = (\theta_1, ..., \theta_n)$, $\theta_i \in \mathbb{R}^p$. Let $G \sim \Pi(\alpha, H)$ be a realisation of a Dirichlet Process (DP) and let $\theta_i | G, \alpha, H \sim G$ independently for $i = 1, ..., n$.

   (i) Write down the definition of the DP in terms of the Dirichlet distribution.

   (ii) Let $A \subseteq \mathbb{R}^p$ and $H(A) = \int_A h(x)dx$. Show that if $G \sim \Pi(\alpha, H)$ then $E(G(A)) = H(A)$.
   Hint: if $X \sim Beta(u, v)$ then $E(X) = u/(u + v)$.

   (iii) In the Blackwell-MacQueen urn process realising $\theta = (\theta_1, \ldots, \theta_n)$,

$$\theta_i | \theta_{<i}, \alpha, H \sim \sum_{j=1}^{i-1} \frac{1}{\alpha + i - 1} \delta_{\theta_j} + \frac{\alpha}{\alpha + i - 1} H,$$

   for $i = 1, \ldots, n$, so $\theta_i \sim H$ is a random draw from H with probability $\alpha/(\alpha + i - 1)$ and otherwise θ_i is chosen randomly from $\theta_{<i} = (\theta_1, \ldots, \theta_{i-1})$. Let $\theta^* = (\theta_1^*, ..., \theta_K^*)$ give the unique values in θ and let $S = (S_1, ..., S_K)$ be a partition of $[n] = \{1, 2, ..., n\}$ with $S_k = \{i \in [n] : \theta_i = \theta_k^*\}$ for $k \in [K]$.
   Let $\pi(\theta^*, S) = \pi(\theta^* | S) P_{\alpha,[n]}(S)$. Show that $\pi(\theta^* | S) = \prod_{k=1}^K h(\theta_k^*)$ and

$$P_{\alpha,[n]}(S) = \alpha^K \prod_{k=1}^K (n_k - 1)! \prod_{i=1}^n \frac{1}{\alpha + i - 1},$$

   where $n_k = |S_k|$ is the number of elements in S_k, $k \in [K]$.
   Hint: the Blackwell-MacQueen process is equivalently a Chinese Restaurant Process.

   (iv) Let $n = 100$. Calculate the prior probability $\theta_1 = \theta_{50} = \theta_{100}$.

(b) [13 marks] Suppose the data $y_b = (y_{b,1}, \ldots, y_{b,n_b})$, $b = 1, \ldots, B$ come in B blocks. Consider a hierarchical Dirichlet process model for these data. Let α and H be given as above and let $G_0 \sim \Pi(\alpha, H)$. Let

$$G_b | G_0 \sim \Pi(\beta, G_0), \quad b \in [B],$$

be B independent realisations of a DP with parameter $\beta > 0$ and base distribution G_0. For $i \in [n_b]$ let $\theta_{b,i} | G_b \sim G_b$ be conditionally independent given G_b (and notice $\theta_{b,i} | G_b, G_0 \sim \theta_{b,i} | G_b$).

   (i) Show that marginally $\theta_{b,i} | G_0 \sim G_0$ for each $b \in [B]$ and $i \in [n_b]$ and hence show that the atoms of G_b are a subset of the atoms of G_0.
   Hint: Assume that G_b and G_0 are discrete distributions.

   (ii) For $b \in [B]$ let $S_b = (S_{b,1}, \ldots, S_{b,K_b})$ be a partition of $\{b\} \times [n_b]$ and let

$$\mathcal{T} = \bigcup_{b \in [B]} \{b\} \times [K_b],$$

   be the set of all pairs (b, k), $b \in [B]$ and $k \in [K_b]$. For each $(b, k) \in \mathcal{T}$ let $\theta_{b,k}^* \sim G_0$. Let $T = (T_1, \ldots, T_L)$ be a partition of $\mathcal{T}$ and for $\ell \in [L]$ let $\theta_\ell^{**} \sim H$. If for some $(b, k) \in \mathcal{T}$ it holds that $(b, i) \in S_{b,k}$ and $(b, k) \in T_\ell$ then $\theta_{b,i} = \theta_{b,k}^* = \theta_\ell^{**}$.
   Calculate the prior $\pi(\theta^{**}, S, T)$ for the random partitions and parameters.

   (iii) Suppose $B = 2$ and $n_1 = n_2 = 100$. What is the probability for $\theta_{1,1} = \theta_{1,50} = \theta_{2,100}$?

   (iv) The observation model is $y_{b,i} \sim p(\cdot | \theta_{b,i})$ jointly independent for $b \in [B]$ and $i \in [n_b]$. Give the posterior $\pi(\theta^{**}, S, T | y)$ up to an overall normalising constant.